DATA ANALYTICS FOR BUSINESS

with Lizhen Xu, Ph.D.

Binary Response Models: Linear Probability Models

Learning Objectives

- Describe binary response problems
- Set up linear probability models
- Define partial effects
- Identify the limitations of linear probability models
- Evaluate the performance of linear probability models

Introduction

- **Regression vs Classification**
  - *Regression*: numeric / quantitative response (continuous / discrete)
  - *Classification*: categorical / qualitative response (nominal / ordinal)
  - Predicting the probability of each category, classification problems can also be handled with regression methods

- **Binary Response:**
  - Dependent variable y can only take two values: 0 or 1
  - Examples:
    - Success" $(y = 1)$ or "Failure" $(y = 0)$
    - Whether to be admitted into a program
    - Whether to default on a loan
    - Whether to participate in a promotion
    - Whether to churn
    - ……

Linear Probability Model (LPM)

$$P(y = 1|X) = \beta_0 + \beta_1 \cdot x_1 + \cdots + \beta_k \cdot x_k$$

- Given the binary nature of y

$$\begin{aligned} E(y|X) &= P(y = 1|X) \cdot 1 + P(y = 0|X) \cdot 0 \\ &= P(y = 1|X) \\ &= \beta_0 + \beta_1 \cdot x_1 + \cdots + \beta_k \cdot x_k \end{aligned}$$

- We can estimate LPM using Ordinary Least Square (OLS)
  - Just like the usual multiple linear regression
  - Regress y (which now takes binary values) on all the x variables
  - The estimated equation:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_k x_k$$

  - The fitted $\hat{y}$ is the predicted probability of "success"

Partial Effects

- Advantages of LPM:
  - Easy to estimate
  - Easy to interpret

- The **partial effect** (or **marginal effect**):
  - How (and by how much) change in one particular explanatory variable x_j will cause change in the *expected* value of y, holding all the other variables unchanged
  - Ceteris paribus
  - Defined as $\frac{\partial E(y|X)}{\partial x_j}$

- For LPM:

$$\frac{\partial E(y|X)}{\partial x_j} = \beta_j$$

  - The coefficient of a variable directly equals the partial effect of that variable

LPM Limitations

- The predicted probability of success, $\hat{y}$, can go outside of the valid range [0,1]
  - Doesn't make sense for some observations
- The partial effects, β_j, stay constant; do not change with the x variable values
- In general, LPM works well for independent variables near the averages in the sample; doesn't work well extreme x values

Model Evaluation

- Evaluate the model: Goodness-of-Fit
  - The usual **R-Squared**
  - For binary response models: **Percent Correctly Predicted**

- **Percent Correctly Predicted**
  - Predict the binary outcome $\tilde{y}_i = 1$, if the estimated prob. $\hat{y}_i \geq c$
  - Predict the binary outcome $\tilde{y}_i = 0$, if the estimated prob. $\hat{y}_i < c$

  - *What threshold c to choose?*
    - A common choice: $c = 0.5$
    - $c = \bar{y}$, the fraction of "success" in the sample
    - Choose c such that the fraction of $\tilde{y}_i = 1$ is the same as $\bar{y}$

Percent Correctly Predicted

- Compare the predicted outcome $\tilde{y}$ against the actual observed outcome y, and construct the confusion matrix

Predicted $\tilde{y}$	Actual y	
	1 (Positive)	0 (Negative)
1 (Positive)	True Pos. (TP)	False Pos. (FP)
0 (Negative)	False Neg. (FN)	True Neg. (TN)

TP: # of obs. w/ $(y = 1, \tilde{y} = 1)$
TN: # of obs. w/ $(y = 0, \tilde{y} = 0)$
FP: # of obs. w/ $(y = 0, \tilde{y} = 1)$
FN: # of obs. w/ $(y = 1, \tilde{y} = 0)$

- The percent correctly predicted (PCP):

$$\frac{TP + TN}{TP + TN + FP + FN}$$

Percent Correctly Predicted (Cont'd)

- High overall PCP doesn't necessarily guarantee good model performance

- Need to check PCP for each possible outcome

$$\frac{TP}{TP+FN} \qquad \frac{TN}{TN+FP}$$

- A good model should be able to predict the least likely outcome also well.

	$y = 1$	$y = 0$
$\tilde{y} = 1$	90	10
$\tilde{y} = 0$	0	0

Summary

- Binary response model setup
- Linear probability model setup and estimation
- Partial effects of an x variable
- Limitations of linear probability models
- Evaluating goodness-of-fit of linear probability models

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Binary Response Models: Logit and Probit Models

Learning Objectives

- Describe latent variable model framework
- Set up Logit and Probit models
- Estimate Logit/Probit models
- Evaluate the performance of Logit/Probit models
- Interpret the coefficients in Logit/Probit models

Latent Variable Models

- Latent Variable Model:

$$y^* = \boldsymbol{\beta_0} + \boldsymbol{\beta_1 x_1} + \cdots + \boldsymbol{\beta_k x_k} + e \quad \equiv \boldsymbol{X\beta} + e$$

$$y = \begin{cases} 1 & \text{if } y^* > 0 \\ 0 & \text{if } y^* \leq 0 \end{cases}$$

  - y^*: latent (unobserved) variable, linearly depending on x
  - y: observed binary outcome, determined by whether $y^* > 0$
  - e: unobserved error term, can be assumed to follow various distributions
  - $\boldsymbol{X\beta} = (1, x_1, \ldots, x_k) \cdot (\beta_0, \beta_1, \ldots, \beta_k)' = \boldsymbol{\beta_0} + \boldsymbol{\beta_1 x_1} + \cdots + \boldsymbol{\beta_k x_k}$: *vectorized*

$$\begin{aligned} \boldsymbol{P(y = 1|X)} &= P(y^* > 0|X) = P(X\beta + e > 0) \\ &= P(e > -X\beta) = 1 - G(-X\beta) \\ &= \boldsymbol{G(X\beta)} \end{aligned}$$

  - $G(\cdot)$ is the ***cdf*** of the distribution of the error term e
    - Assumed to be **symmetric** such that $1 - G(-X\beta) = G(X\beta)$

Logit Model

- e follows the standard **logistic distribution**
- Cumulative distribution function (cdf)

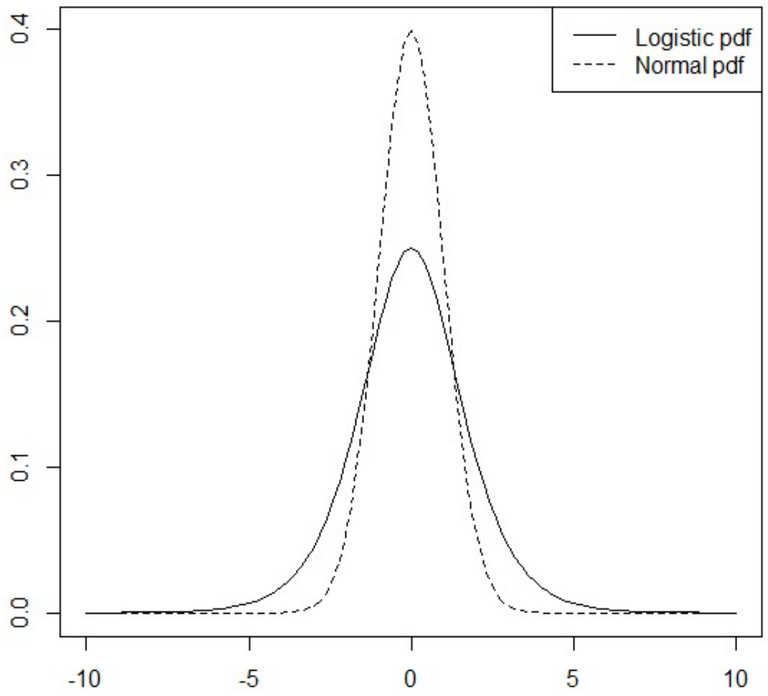
$$G(x) = \frac{e^x}{1 + e^x}$$

- Probability density function (pdf)

$$g(x) = \frac{dG(x)}{dx} = \frac{e^x}{(1 + e^x)^2}$$

- <u>Note</u>: logistic distribution is <u>*symmetric*</u>

$$g(x) = g(-x)$$



- The probability of success:

$$\boldsymbol{p(y = 1|X) = G(X\beta) = \frac{e^{X\beta}}{1 + e^{X\beta}} = \frac{\exp(\beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k)}{1 + \exp(\beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k)}}$$

Probit Model

- e follows the standard **normal distribution**
- Probability density function (pdf)

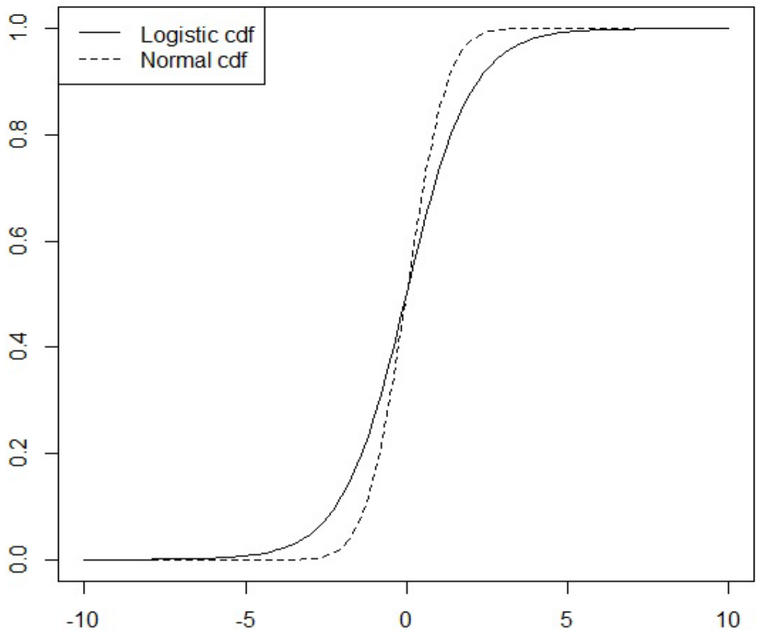
$$g(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

- Cumulative distribution function (cdf)

$$G(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz$$

- The probability of success:

$$\boldsymbol{p(y = 1|X) = G(X\beta) = \int_{-\infty}^{X\beta} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz}$$



Estimating Logit/Probit Models

- Logit/Probit models can be estimated by **maximum likelihood estimation (MLE)**

$$\hat{\beta} = \underset{\beta}{\operatorname{argmax}}\, \mathcal{L}(\beta)$$

- Likelihood function:

$$\mathcal{L}(\beta) = \prod_{i:y_i=1} G(X_i\beta) \prod_{i':y_{i'}=0} (1 - G(X_i\beta))$$

- $\hat{p}(y_i = 1|X_i)$ as close to 1 as possible for those with $y_i = 1$
  $\hat{p}(y_i = 1|X_i)$ as close to 0 as possible for those with $y_i = 0$

Model Evaluation

- Goodness of Fit:
  - Usual R-Squared
  - AIC, BIC
  - Percent Correctly Predicted
    - Predict $\tilde{y}_i = 1$, if $G(X_i\hat{\beta}) \geq c$
    - Predict $\tilde{y}_i = 0$, if $G\left(X_i\hat{\beta}\right) < c$
    - Choice of threshold c
    - Confusion matrix
    - Good model needs to be able to predict the least likely outcome

Model Interpretation: Log-Odds

- For Logit Models:

$$\boldsymbol{p(y=1|X) = \frac{\exp(\beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k)}{1 + \exp(\beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k)}}$$

  - Given X value, we can easily predict the probability of success

- We can also re-write the formula:

$$\frac{p(y=1|X)}{1-p(y=1|X)} = \exp(X\beta)$$

$$\ln\left(\frac{p(y=1|X)}{1-p(y=1|X)}\right) = X\beta$$

- ***Odds***: $\frac{p(y=1|X)}{1-p(y=1|X)}$
  - Traditionally used in horse-racing
  - β_j: one-unit increase in x_j changes log-odds by β_j, or multiplies the odds by e^{β_j}

Model Interpretation: Partial Effects

- Interpreting the Logit/Probit Estimates
  - $P(y = 1|X) = G(X\beta)$: non-linear (in model parameters)
  - The coefficients β no longer have direct interpretation

- Partial (marginal) Effects

$$E(y|X) = P(y = 1|X) = G(X\beta)$$

$$\frac{\partial E(y|X)}{\partial x_j} = \frac{\partial G(X\beta)}{\partial x_j} = g(X\beta) \cdot \beta_j$$

Model Interpretation: Partial Effects (Cont'd)

$$\frac{\partial E(y|X)}{\partial x_j} = g(X\beta) \cdot \beta_j$$

- The coefficient β_j has the same sign as the partial effect
- The magnitude of the partial effect equals the coefficient multiplied by a factor, $g(X\beta)$, the pdf of the error term evaluated at the X values, $(x_1, \ldots, x_k)$.
- The partial effects *change* with the explanatory variable values!
- Partial effects are comparable across different models;

  Coefficients are NOT --- they need to be scaled by different factors
- The partial effects are the *greatest* when $\boldsymbol{X\beta}$ is <u>around zero</u>;

  The partial effects are the *weakest* when $\boldsymbol{X\beta}$ is <u>very large or very small</u>
- $P(y = 1|X)$ is *S-shaped*

Summary

- Latent variable model framework
- Logit and Probit model setup and assumptions
- Logit/Probit model estimation and evaluation
- Interpretation of Logit/Probit models
  - Log-odds
  - Partial (marginal) effects